

[Collaborative Docs](#) >

[Paper 6] Paper "HighresRun" & "Clumps" - 1e12q High-Res Run To Study Clumps

This page presents the outline, ongoing efforts, results, and future plans for the AGORA Paper "Clumps" (being co-led by Ceverino, Lupi, etc).

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Important recent changes that require your attention are in red color.

(last revision: 08/14/2019)

[1. Project Outline](#)

	Paper "HighresRun" + Paper "Clumps"																					
GOALS	• Paper "HighresRun" : Run ~1e12 Ms halo at higher resolution (20 pc instead of 80 pc) to study SF clumps at high z, nice resolution study when compared to Paper "CosmoRun" • Paper "Clumps" : Study star-forming clumps, compare with observations																					
LEADERS	• Daniel Ceverino (U. Heidelberg) • Alessandro Lupi (IAP)																					
PARTICIPANTS	• Interested code representatives: <table><tr><th>Code</th><th>Contacts</th></tr><tr><td>ART-I</td><td>Ceverino, Dekel, Lee, Mandelker, Primack, Roca-Fabrega</td></tr><tr><td>ART-II</td><td>(Feldmann)</td></tr><tr><td>CHANGA</td><td>(Quinn)</td></tr><tr><td>ENZO</td><td>Kim (Ji-hoon), O'Shea</td></tr><tr><td>GADGET</td><td>Nagamine, Shimizu</td></tr><tr><td>GASOLINE</td><td>(Mayer, Wadsley)</td></tr><tr><td>GEAR</td><td>Hausammann, Revaz</td></tr><tr><td>GIZMO</td><td>Dong, Hopkins, Lupi ("favorite" fbck is their standard mechanical fbck, GALSF_FB_MECHANICAL in here, *not* FIRE-2)</td></tr><tr><td>RAMSES</td><td>Agertz, Roca-Fabrega</td></tr></table>		Code	Contacts	ART-I	Ceverino , Dekel, Lee, Mandelker, Primack, Roca-Fabrega	ART-II	(Feldmann)	CHANGA	(Quinn)	ENZO	Kim (Ji-hoon) , O'Shea	GADGET	Nagamine , Shimizu	GASOLINE	(Mayer, Wadsley)	GEAR	Hausammann , Revaz	GIZMO	Dong, Hopkins, Lupi ("favorite" fbck is their standard mechanical fbck, GALSF_FB_MECHANICAL in here , *not* FIRE-2)	RAMSES	Agertz, Roca-Fabrega
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	(Code leaders as of Aug. 2019 / ICs help: Buehlmann, Hahn / Analysis help: Chakrabarti, Kim, Turk) • Working Groups engaged: - III (common cosmo ICs) - IV (common analysis) - XI (High-redshift Galaxies)																					
SCIENCE CASE	• Observationally, most star-forming galaxies have off-center clumps (e.g., Guo et al. 2015). • Can we resolve recent conflicting results by ART-I, GASOLINE/ChaNGa, GIZMO? • Numerical issues such as viscosity, dissipation, non-conservation of angular momentum could be discussed. The topic is still impacted by numerical implementation of the hydro as much as subgrid physics. • A related topic is secular disk instabilities. How do different hydro codes (and their gravity solvers) generate bulges of spiral galaxies and impact galactic morphology? • What do we learn from Mayer's latest paper and Mayer/Lupi/Wadsley's earlier paper? Is the difference driven mostly																					

by differences in feedback or in hydro schemes?

• Pre-production discussion/to-do's:

PRE-PRODUCTION
DISCUSSION

1ST MEETING
(04/24/2017)

2ND MEETING
(07/06/2017)

3RD MEETING
(08/12/2017)

4TH MEETING
(06/11/2018)

5TH MEETING
(08/11/2018)

6TH MEETING
(08/10/2019)

Item	To-do
Initial condition	• As of 06/2018, we are switching to use the AGORA quiescent 10^{12} Ms halo instead of the 10^{13} Ms halo. Therefore, the description about the AGORA 10^{13} Ms halo (gray color below) is outdated. For more information about the 10^{12} Ms halo, see Paper "CosmoRun"+"CGM". A higher-resolution IC with slightly reduced Lagrangian volume targeted for $z=1$ will be produced by Buehlmann and Hahn (08/2019). • Also, as of 06/2018, to best utilize the common features in simulations for Papers "CGM" and "Clumps", and the time of the code leaders, we strategically decided to tackle Paper "CosmoRun"+"CGM" first, then Paper "HighresRun"+"Clumps". • We may later simulate 10^{13} Ms halo as well, which is described below. This would be considered if the 10^{12} Ms halo turns out to be not so interesting with its small progenitor mass at $z=2$ (when we think the formation of giant clumps are at its peak). In this case, comparison should be made at $z > 2-3$ before AGN becomes important (see below). • We may run and compare the 10^{13} Ms halo at $z=0$. (More information about the halo at this link.) The 10^{13} Ms halo will produce $\sim 1.2 \times 10^{12}$ Ms halo at $z=2$. The target resolution at $z=2$ will be ~ 100 pc. • The convex hull ICs for the 10^{13} Ms halo can be found in Fiacconi et al. 2016 or Fiacconi et al. 2017 need to be modified for a target redshift of 2 (Fiacconi et al. ICs were for target redshifts of 0 and 6, respectively). • In the meantime, we will tentatively use an ellipsoidal IC. A MUSIC configuration file for an ellipsoidal IC with levelmax=12 and a Lagrangian region of $4R_{\text{vir}}$ at a target redshift 2 is created by Buehlmann and Hahn ($z=2$ snapshot of a DM-only test with levelmax=11, comparison of IC regions). • In fact, they now have built a full library of AGORA ICs with different choices of Lagrangian regions for progenitors at different redshifts (to be formally announced soon; example plot).
Target redshift	• $z=1$ for a 10^{12} Ms halo • $z=2$ for a 10^{13} Ms halo (if we later decide to run this IC)
Definition of a "clump"	• To simplify our comparison, we focus only on the "giant" clumps in a high-z, high-f_{gas} galaxy (i.e., each clump of mass more than 1–2% of the total disk mass, or, each clump with SFR more than 10% of the total disk SFR). • By focusing on "giant" clumps, we can prevent the result of our comparison from depending too much on hard-to-calibrate parameters (e.g., pressure floor). For these "giant" clumps, there also exist good observational counterparts (e.g., Guo et al.). • This means that we do not focus on small "2nd-order" clumps, e.g., small clumps along galactic spiral arms in a low-f_{gas} galaxy.
	• As of Jun. 2018, the following calibration approach is on hold, as we combine the efforts to launch Papers "CGM" and "Clumps". See the calibration steps described in Paper "CosmoRun"+"CGM". • However, there remains a concern that a separate feedback calibration step will be needed as we choose to adopt a higher-resolution IC for Paper "HighresRun"+"Clumps" (i.e., 20 pc resolution instead of 80 pc; raised by Ceverino). • The "favorite" feedback prescriptions will need to be first calibrated across codes in a "massive gas disk" regime to produce similar properties (e.g., M_*/M_{halo} ratios). For this, we will carry out a 2-step calibration as follows: – (1) Step #1: We will first run the isolated disks similar to our disk comparison paper (formerly Paper "4"; its Workspace) but with massive gas disks (50% gas fraction). Agertz provided a new isolated disk IC with 50% gas fraction in this Dropbox link. We will run this IC in three modes (Grackle cooling only, "common" feedback identical to our disk comparison

Common physics (1): Stellar feedback	paper, and "favorite" feedback). We will have a 0.2 Gyr relaxation step (adiabatic run) before the cooling takes place in order to reduce the transient phase of catastrophic star formation). Each group will tweak the "favorite" recipe to agree on the M_*/M_{halo} ratio (number to be chosen) at 0.5 Gyr. You are encouraged to talk to the person who ran the simulation for the disk comparison paper, to properly read in the IC, and acquire the parameter set we settled on only after multiple trials and errors. – (2) Step #2: Then we will use the "pre-calibrated" "favorite" feedback to run the cosmological IC, and compare the M_*/M_{halo} ratios across codes at $z=4$ (and then at $z=2$?) to overcome the overcooling problem. Hopefully, the pre-calibration in the isolated disk will help us to acquire similar stellar masses. If not, we will probably have to carry out another calibration in a cosmological setting. • ART-I (Ceverino et al.) will use kinetic and thermal feedback modes plus radiation pressure. • Our prediction based on e.g., Ceverino's earlier experience or the Scylla comparison is that we don't have to worry too much about this calibration step.
Common physics (2): AGN feedback	• We expect that AGN feedback will not be needed as long as we compare galaxies at $z=2$ or earlier. • But in order to compare the $1e13 M_{\odot}$ halo at $z<2$, we will need to consider AGN feedback.
Common physics (3): Pressure floor	• We will simply adopt the prescription used in our disk comparison paper (formerly Paper "4"), because the "giant" clumps do not depend on the pressure floor, but mostly just on Toomre Q.
Common physics (4): Star formation	• We will start from the prescription used in our disk comparison paper (formerly Paper "4"), and see if we need to modify it.
Common physics (5): Cooling & UVB	• We will adopt the Grackle equilibrium cooling used in our disk comparison paper (formerly Paper "4") with self-shielding (see the Paper "CosmoRun"+"CGM" Workspace for more information). • Ceverino will check if the physics and metallicity definitions are consistent between different versions of Grackle adopted in different codes. For more discussion on the metallicity definitions in different versions of Grackle, search "[Note 6]" in the Paper "Disk" Workspace.
Code-dependent physics: Metal diffusion in SPH	• Each group will decide what they want to do. For example, GASOLINE/CHANGA/GIZMO have the explicit metal diffusion scheme, while GADGET/GEAR(?) do not have it.
Softening, smoothing, and refinement strategy	• We will impose 4 times the resolution used in Paper "CosmoRun"+"CGM", i.e., 20 proper pc for SPH, 41 comoving pc for AMR, with an IC of a smaller Lagrangian volume (see the "Initial condition" above). • This will make a great case for Paper "HighresRun" in which we carry out a resolution study by comparing with Paper "CosmoRun".
Output strategy	• See this link. But you are encouraged to store more snapshots on our large NERSC disk space, in order to investigate e.g., clump's lifetime.
Common analysis	• We may begin from common analysis routines in agora-analysis-script used for our disk comparison paper and build upon it. • Mandelker volunteers to incorporate his clump finding routine (originally in fortran) into the agora-analysis-script repository. • Huertas-Company and Primack may incorporate the clump finding routine using machine learning (developed by Lee and Huertas-Company, and being used by Ginzburg and Dekel) into our common analysis script. • We may also build a gas/stellar clump finder in our common analysis script (e.g., simply identifying the high density peaks in Ceverino's previous studies). • Joel Primack, Raymond Simons, Gregory Snyder may help us to utilize the yt+Sunrise pipeline to produce realistic images so we can apply procedures used by observers. If done, this would allow us to use the exact same clump finding routine as in Guo et al. Will compare with observations. • We may consider identifying and studying additional structures related to gravitational instability (e.g., inflow on the disk to the galactic center, bulges, spirals, etc.)
Key plots to make	• Face-on surface density • Clump mass function • Gradients of stellar age, specific star formation rate • Toomre Q maps • Sunrise images

		• ex-situ vs. in-situ clumps, clump migration • Disks and vertical structures (especially when 20 pc resolution is reached) • Emission lines
	Obs. counterpart	• e.g., Guo et al. 2015
TIMELINE AND MILESTONES	• Before and during the Aug. 2017, Aug. 2018 Workshop: Finalize runtime parameters and simulation strategy. • Shortly after the Aug. 2019 Workshop: End of calibration steps in Paper "CosmoRun"+"CGM". • By 12/2019: Monitor the emergence of clumps in the production runs of Paper "CosmoRun". If and when decided, start the "HighresRun" simulation with a high-resolution IC with a smaller Lagrangian region. • As soon as we have the production simulations at $z=2-1$: First analysis + discussion + paper writing (Overleaf link below). Online meetings to be scheduled.	
OTHER LINKS	• WG XI Workspace (High-z Galaxies)	

2. Results

Below we post the first batch of composite figures we agreed to produce for Paper "Clumps". This is hardly the final result of the comparison, so note that some figures may change significantly as we identify and fix various issues. Reload the images to make sure you are looking at the most recent ones. Please also read Code Notes below carefully before drawing any conclusion. We also kindly remind you of the [internal policy](#) regarding the disclosure of AGORA simulation results.

(1) Paper "HighresRun" and Paper "Clumps" Draft

- Overleaf link: <https://www.overleaf.com/18564401zbqjgzfgyvbw>

(2) Calibration of Feedback Prescriptions

See the result at the [Paper "CosmorRun"+"CGM" Workspace](#).

(3) Production Runs and Paper Draft

	Data Locations, Visualizations, Metrics to Compare